

Figure 1: Reference architecture for edge computing

IoT communication/transport layer protocols

The following are some of the widely used IoT communication/transport layer protocols.

- **Bluetooth:** This is a wireless technology for data exchange between electronic devices over short distances. An electronic device must meet Bluetooth support standards to enable communication over Bluetooth protocols.
- **Wi-Fi:** This is a wireless local area networking technology. It is a very common and easy-to-set-up communication/transport layer. Any electronic devices within range of a wireless modem can attempt to access the network.
- **NFC:** Near Field Communication (NFC) is a protocol to enable two electronic devices to establish communication. It is used for contactless payments, electronic tickets, mobile payments, sharing contacts, photos, videos, files, etc. It offers simple and safe two-way interaction between electronic devices within a specified range.
- **Zigbee:** This is a high-level communication protocol for personal area networks. Zigbee is a low power, low data rate and close proximity wireless network. It is simpler and less expensive than Bluetooth and Wi-Fi. It is used for home automation, medical devices, traffic management systems and other consumer or industrial low power, low bandwidth requirements for small scale projects.
- **LoRaWAN:** Low Power Wide Area Network (LPWAN/LoRaWAN) operates in the radio spectrum for wide area networks. Similar to Wi-Fi, it can be set up for using lower radio frequencies with a longer range. It allows low-powered devices to communicate with Internet connected applications.

There are some other transport layer protocols too, such as Z-Wave, 6LoWPAN, Thread, Cellular, Ethernet, Eddystone, WiMax, etc.

IoT data protocols

Data protocols are a set of rules for establishing communication between various entities of the system. These protocols define the syntax, semantics, synchronisation of data, and provision for error recovery. The following are some of the widely used IoT data protocols.

- **MQTT:** Message Queue Telemetry Transport (MQTT) is designed to provide embedded connectivity between applications and middleware on the one side, and networks and communications on the other. It follows a publish/

subscribe architecture, where the system consists of three main components: publishers, subscribers and a broker.

- **AMQP:** The Advanced Message Queuing Protocol (AMQP) runs over TCP and provides a publish/subscribe architecture that is similar to MQTT. The difference is that the broker is divided into two main components: the exchange and queues. The exchange is responsible for receiving publisher messages and distributing them to queues based on predefined roles and conditions. Queues basically represent the topics, and its subscribers get the sensory data whenever it is available in the queue.
- **CoAP:** The Constrained Application Protocol (CoAP) is for constrained devices called nodes to communicate with the wider Internet using the same protocols. It is designed to be used under the same constrained communication network between devices and nodes on the Internet. Multi-cast, low overheads and simplicity are important features of CoAP.
- **HTTP:** This is the standard protocol for Web services and is used in IoT solutions. The most popular architectural style, called RESTful, is widely used on mobile and Web applications, and is being considered for IoT solutions.

There are some other data communication protocols such as:

- Mosquitto: An open source MQTT broker
- XMPP (Extensible Messaging and Present Protocol)
- DDS (Data Distribution Service for real-time systems)
- LLAP (Lightweight Local Automation Protocol)
- LWM2M (Lightweight M2M)
- SSI (Simple Sensor Interface)

The edge gateway and middleware gateway

Edge gateways process initial accumulated data that is received from IoT devices/sources and convert that into an expected format. They share the accumulated data from IoT devices to the middleware gateway, which is an IoT and API gateway. The IoT gateway empowers the system to have bi-directional communication between IoT devices and storage/analytics systems. It helps to regulate the environmental changes and detect possible issues with the functioning of a system. These gateways protect information moving in both directions, as well as prevent unauthorised control of IoT devices from the outside world. They also facilitate the device's life cycle management. Figure 2 depicts the various stages of this life cycle and the respective features for device management.

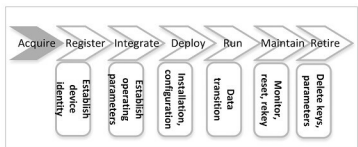


Figure 2: Life cycle management of a device